Converting ER Diagram into Table

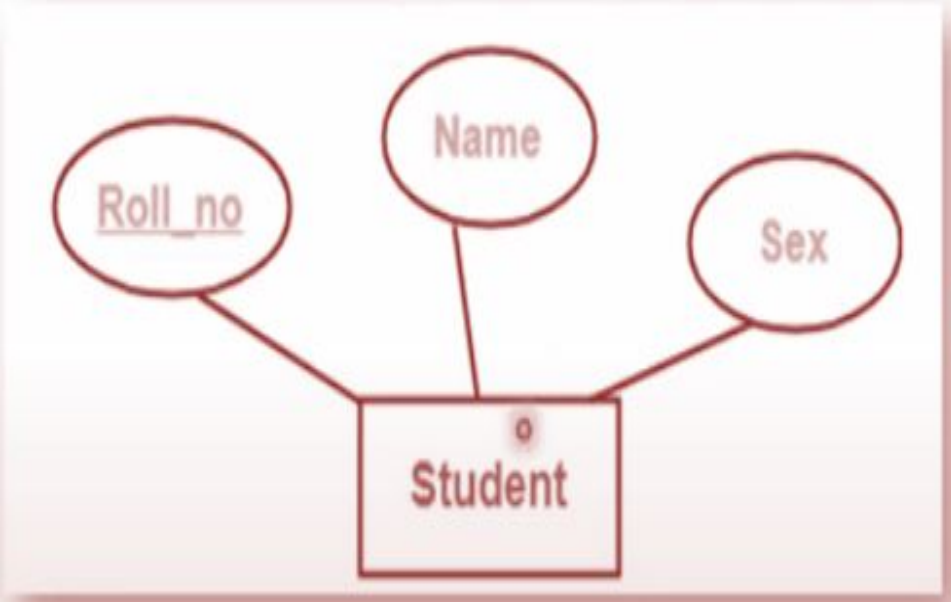
UNIT-3(c)

Steps to map ER diagram to relational schema

Rules of Converting ER Diagram into Table

➢ Rule 1: Strong Entity set with Simple attributes

- Attributes of the table will be the attributes of the entity set.
- The primary key of the table will be the key attribute of the entity set.



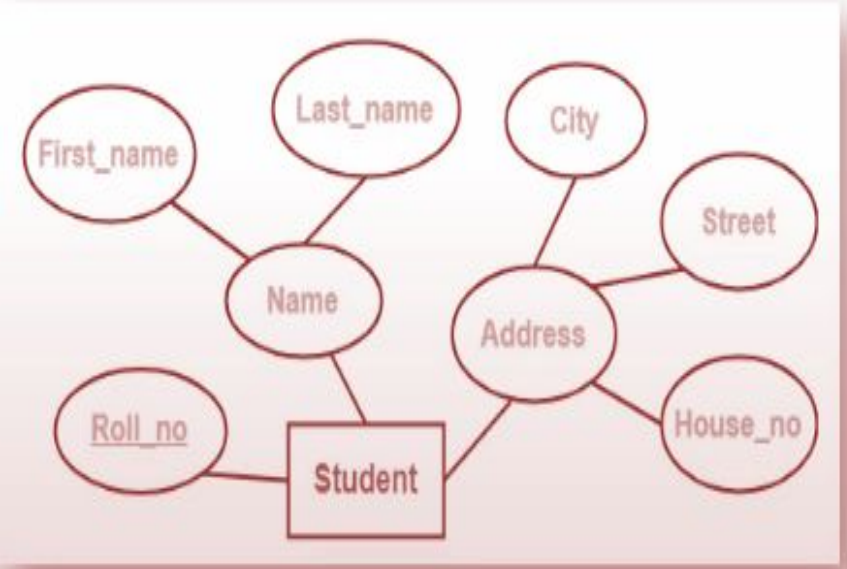
Roll_no	Name	Sex

Schema : Student (Roll_no , Name , Sex)

Rules of Converting ER Diagram into Table

➢Rule 2: For Strong Entity Set With Composite Attributes

- A strong entity set with any number of composite attributes will require only one table in relational model.
- While conversion, simple attributes of the composite attributes are taken into account and not the composite attribute itself.



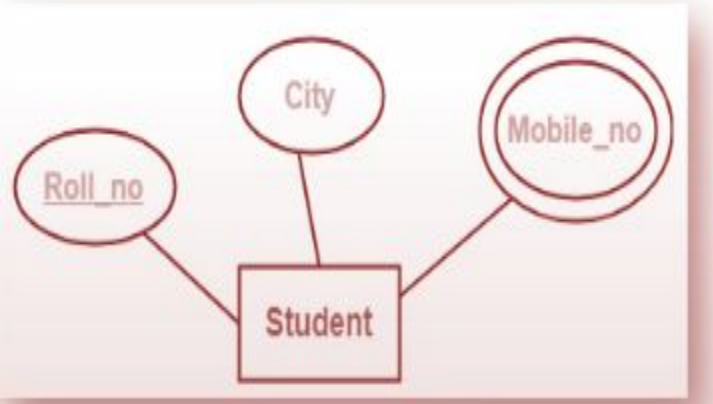
Roll_no	First_name	Last_name	House_no	Street	City

Schema : Student (Roll_no , First_name , Last_name , House_no , Street , City)

Rules of Converting ER Diagram into Table

➢Rule 3: For Strong Entity Set With Multi Valued Attributes

- A strong entity set with any number of multi valued attributes will require two tables in relational model.
- One table will contain all the simple attributes with the primary key.
- Other table will contain the primary key and all the multi valued attributes.



Roll_no	City

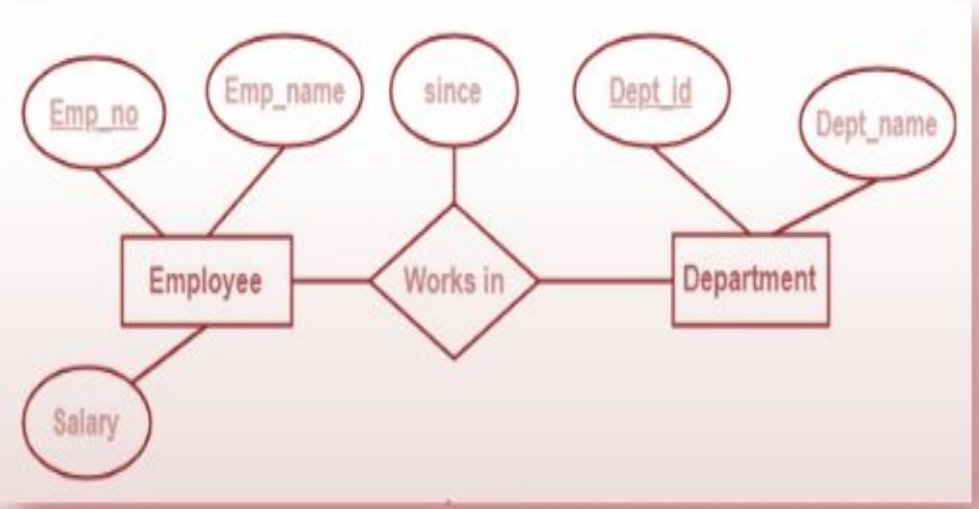
Roll_no	Mobile_no

Rules of Converting ER Diagram into Table

➢ Rule 4: Translating Relationship Set into a Table

Attributes of the table are-

- Primary key attributes of the participating entity sets
- Its own descriptive attributes if any.
- Set of non-descriptive attributes will be the primary key.



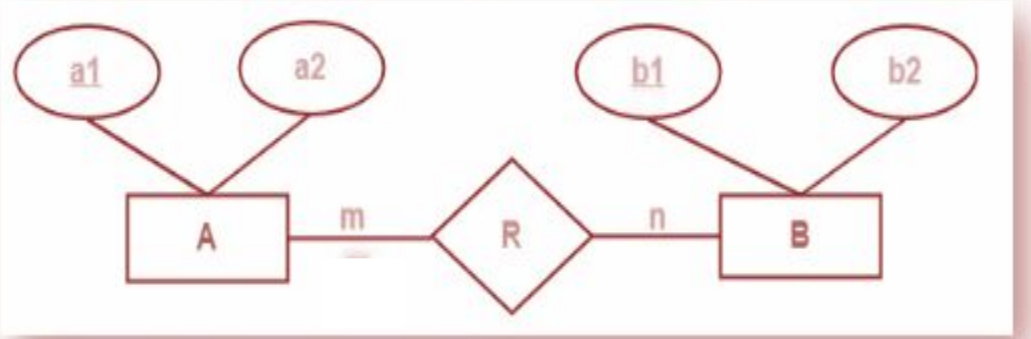
Emp_no	Dept_id	since

Schema : Works in (Emp_no , Dept_id , since)

Rules of Converting ER Diagram into Table

➢ Rule 5: For Binary Relationships With Cardinality Ratios

✓ Case-1: For Binary Relationship With Cardinality Ratio m:n



Here, three tables will be required-

1. A (a1 , a2)
2. R (a1 , b1)
3. B (b1 , b2)

✓ Case-2: For Binary Relationship With Cardinality Ratio 1:n

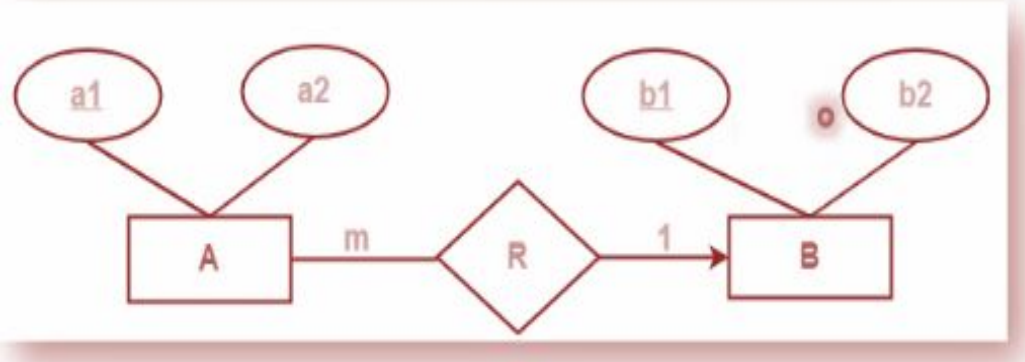
Here, two tables will be required-

1. A (a1 , a2)
2. BR (a1 , b1 , b2)

Rules of Converting ER Diagram into Table

➢ Rule 5: For Binary Relationships With Cardinality Ratios

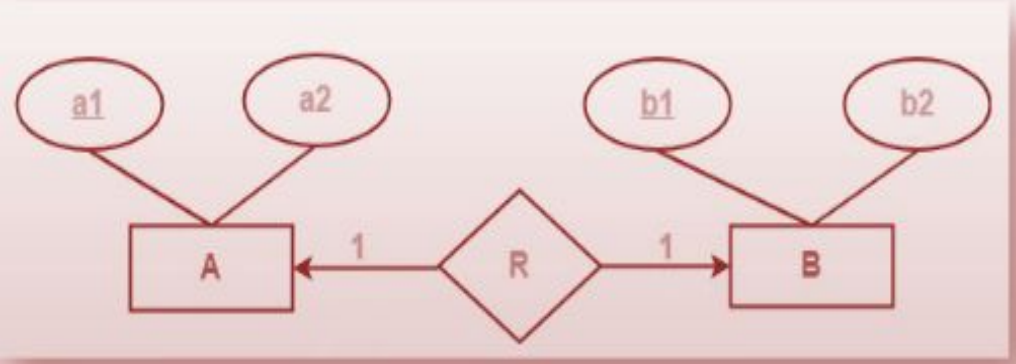
✓ Case-3: For Binary Relationship With Cardinality Ratio m:1



Here, two tables will be required-

1. AR (a1 , a2 , b1)
2. B (b1 , b2)

✓ Case-4: For Binary Relationship With Cardinality Ratio 1:1



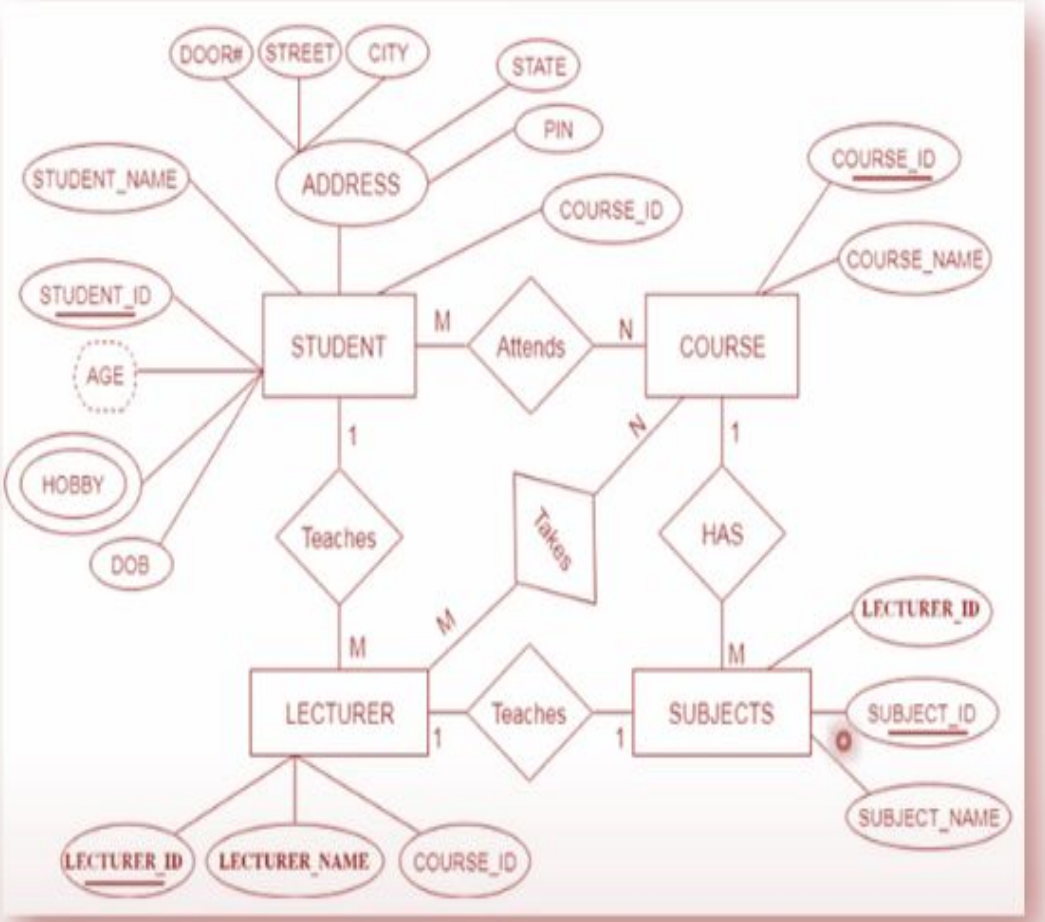
Way-01:

1. AR (a1 , a2 , b1)
2. B (b1 , b2)

Way-02:

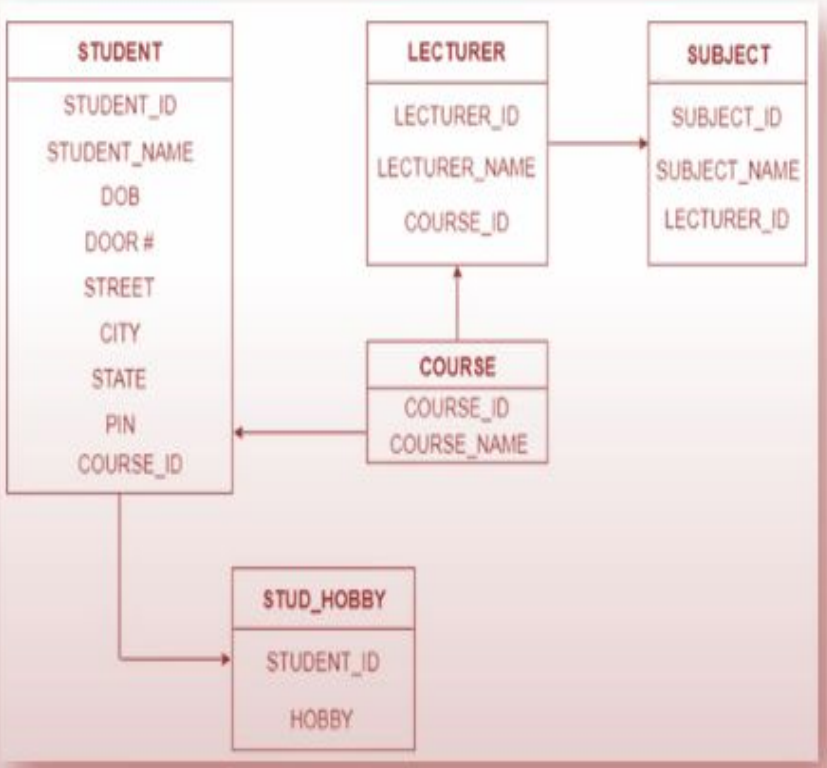
1. A (a1 , a2)
2. BR (a1 , b1 , b2)

Example of Converting ER Diagram into Table

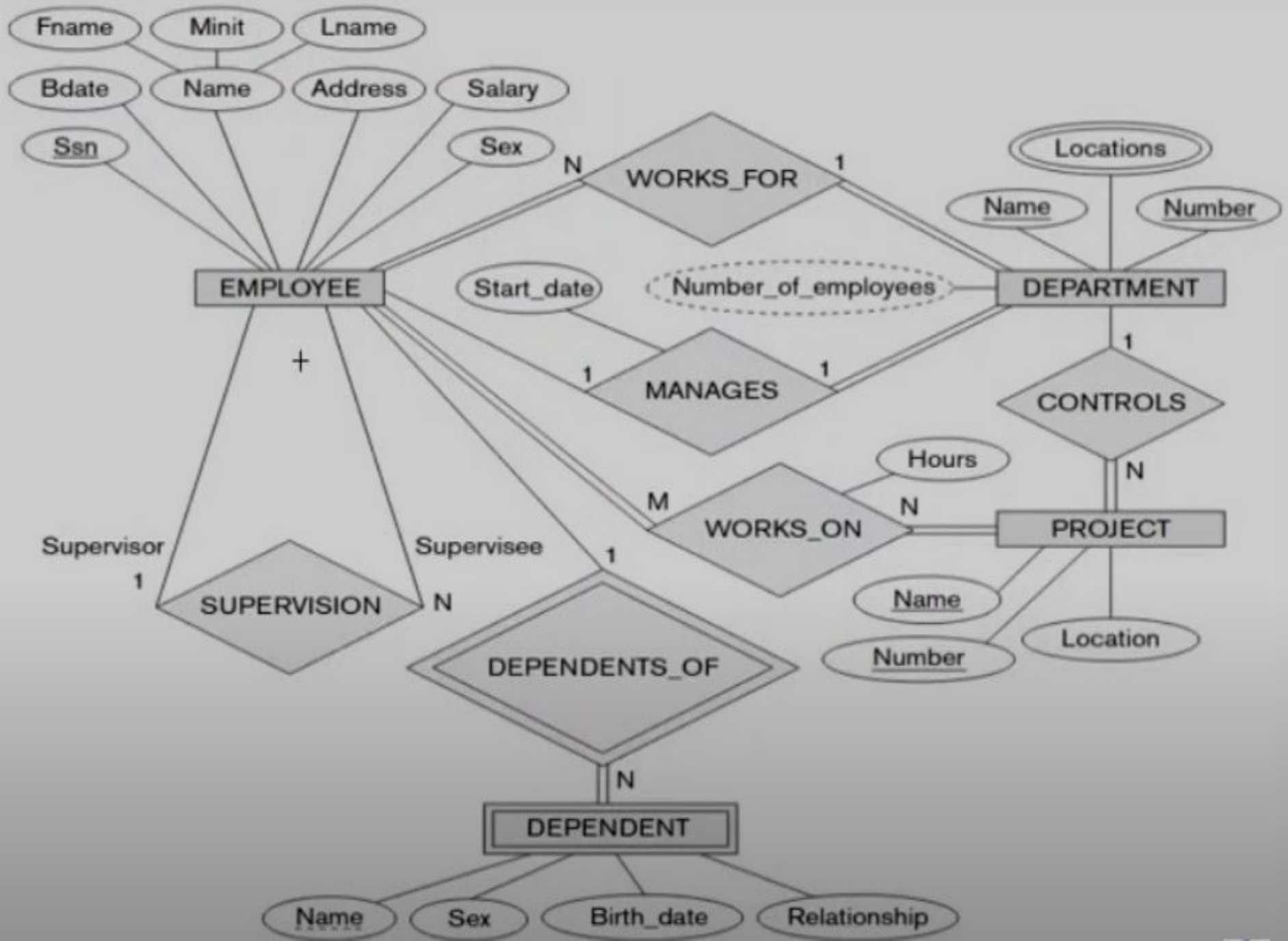


ER Diagram

Table



The ER conceptual schema diagram for the COMPANY database.



ER-to-Relational Mapping Algorithm

- Step 1: Mapping of Regular Entity Types
- Step 2: Mapping of Weak Entity Types
- Step 3: Mapping of Binary 1:1 Relation Types
- Step 4: Mapping of Binary 1:N Relationship Types.
- Step 5: Mapping of Binary M:N Relationship Types.
- Step 6: Mapping of Multivalued attributes.
- Step 7: Mapping of N-ary Relationship Types.

ER-to-Relational Mapping Algorithm

- **Step 1: Mapping of Regular Entity Types.**
  - For each regular (strong) entity type E in the ER diagram, create a relation R that includes all the simple attributes of E.
  - Choose one of the key attributes of E as the primary key for R.
  - If the chosen key of E is composite, the set of simple attributes that form it will together form the primary key of R.
- **Example:** We create the relations EMPLOYEE, DEPARTMENT, and PROJECT in the relational schema corresponding to the regular entities in the ER diagram.
  - SSN, DNUMBER, and PNUMBER are the primary keys for the relations EMPLOYEE, DEPARTMENT, and PROJECT as shown.

ER-to-Relational Mapping Algorithm

Step 1: Mapping of Regular Entity Types.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary

DEPARTMENT

Dname	Dnumber

PROJECT

Pname	Pnumber	Plocation

ER-to-Relational Mapping Algorithm

- **Step 2: Mapping of Weak Entity Types**
  - For each weak entity type W in the ER schema with owner entity type E, create a relation R & include all simple attributes (or simple components of composite attributes) of W as attributes of R.
  - Also, include as foreign key attributes of R the primary key attribute(s) of the relation(s) that correspond to the owner entity type(s).
  - The primary key of R is the *combination of* the primary key(s) of the owner(s) and the partial key of the weak entity type W, if any.
- **Example:** Create the relation DEPENDENT in this step to correspond to the weak entity type DEPENDENT.
  - Include the primary key SSN of the EMPLOYEE relation as a foreign key attribute of DEPENDENT (renamed to ESSN).
  - The primary key of the DEPENDENT relation is the combination {ESSN, DEPENDENT_NAME} because DEPENDENT_NAME is the partial key of DEPENDENT.

ER-to-Relational Mapping Algorithm

Step 2: Mapping of Weak Entity Types

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship

ER-to-Relational Mapping Algorithm

- **Step 3: Mapping of Binary 1:1 Relation Types**

  +

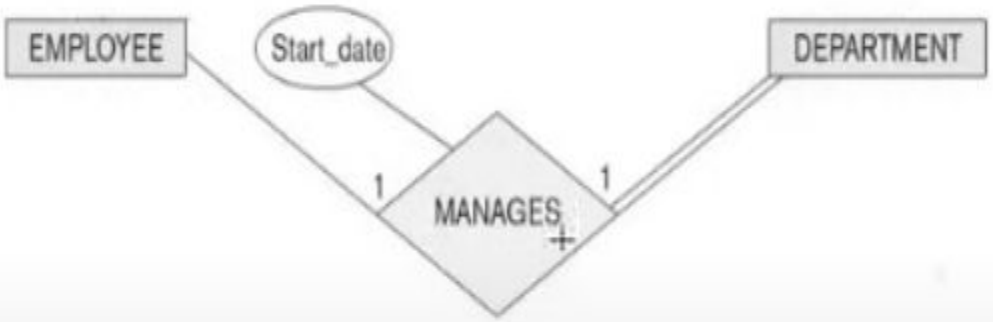
  - For each binary 1:1 relationship type R in the ER schema, identify the relations S and T that correspond to the entity types participating in R.
  - Choose one of the relations-say S-and include a foreign key in S the primary key of T. It is better to choose an entity type with total participation in R in the role of S.
- **Example:** 1:1 relation MANAGES is mapped by choosing the participating entity type DEPARTMENT to serve in the role of S, because its participation in the MANAGES relationship type is total.

ER-to-Relational Mapping Algorithm

Step 3: Mapping of Binary 1:1 Relation Types

DEPARTMENT

Dname	Dnumber



DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date

ER-to-Relational Mapping Algorithm

- **Step 4: Mapping of Binary 1:N Relationship Types.**
  - For each regular binary 1:N relationship type R, identify the relation S that represent the participating entity type at the N-side of the relationship type.
  - Include as foreign key in S the primary key of the relation T that represents the other entity type participating in R.
  - Include any simple attributes of the 1:N relation type as attributes of S.
- **Example:** 1:N relationship types WORKS_FOR, CONTROLS, and SUPERVISION in the figure.
  - For WORKS_FOR we include the primary key DNUMBER of the DEPARTMENT relation as foreign key in the EMPLOYEE relation and call it DNO.

ER-to-Relational Mapping Algorithm

Step 4: Mapping of Binary 1:N Relationship Types.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno

PROJECT

Pname	Pnumber	Plocation

PROJECT

Pname	Pnumber	Plocation	Dnum

ER-to-Relational Mapping Algorithm

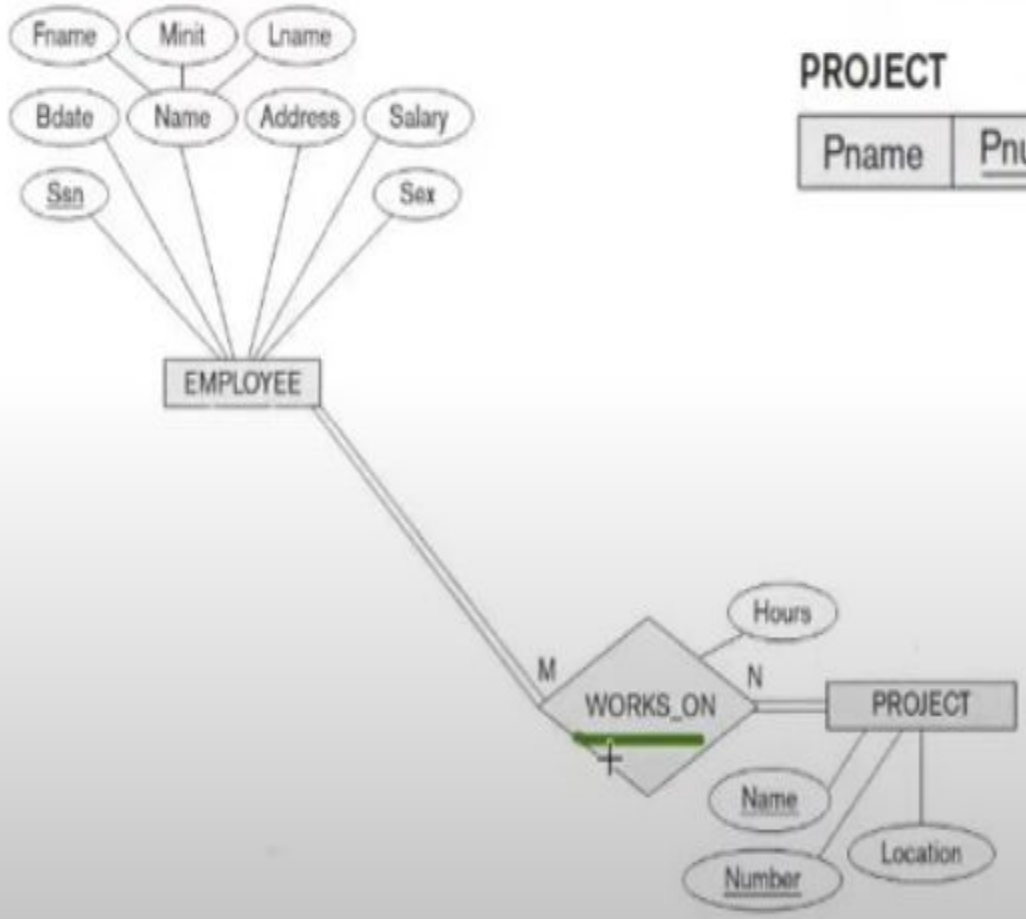
- **Step 5: Mapping of Binary M:N Relationship Types.**
  - For each regular binary M:N relationship type R, *create a new relation* S to represent R.
  - Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types; *their combination will form the primary key* of S.
  - Also include any simple attributes of the M:N relationship type (or simple components of composite attributes) as attributes of S.
- **Example:** The M:N relationship type WORKS_ON from the ER diagram is mapped by creating a relation WORKS_ON in the relational database schema.
  - The primary keys of the PROJECT and EMPLOYEE relations are included as foreign keys in WORKS_ON and renamed PNO and ESSN, respectively.
  - Attribute HOURS in WORKS_ON represents the HOURS attribute of the relation type. The primary key of the WORKS_ON relation is the combination of the foreign key attributes {ESSN, PNO}.

ER-to-Relational Mapping Algorithm

Step 5: Mapping of Binary M:N Relationship Types.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno



PROJECT

Pname	Pnumber	Plocation	Dnum

WORKS_ON

Essn	Pno	Hours

ER-to-Relational Mapping Algorithm

Step 5: Mapping of Binary M:N Relationship Types.

EMPLOYEE

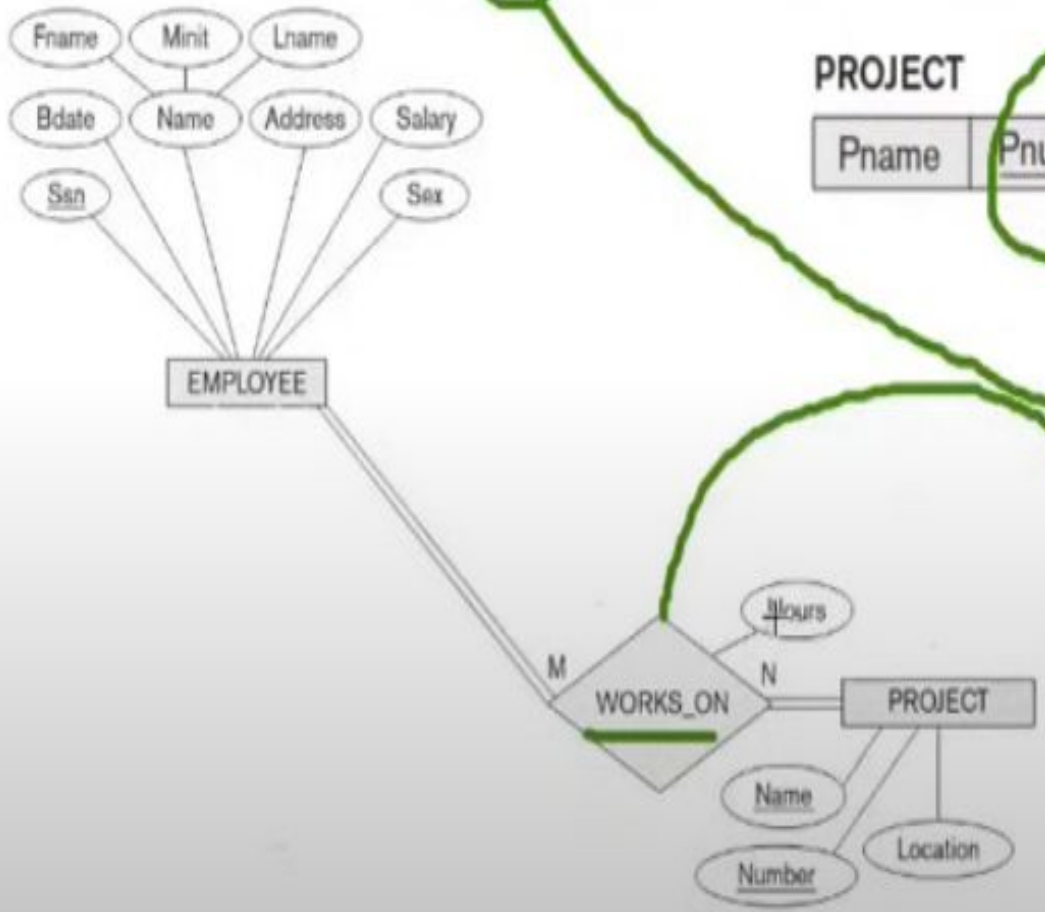
Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno

PROJECT

Pname	Pnumber	Plocation	Dnum

WORKS_ON

Essn	Pno	Hours



ER-to-Relational Mapping Algorithm

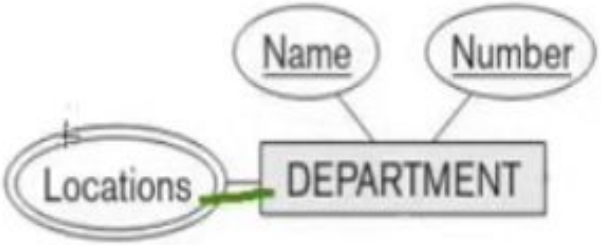
- **Step 6: Mapping of Multivalued attributes.**
  - For each multivalued attribute A, create a new relation R.
  - This relation R will include an attribute corresponding to A, plus the primary key attribute K-as a foreign key in R-of the relation that represents the entity type of relationship type that has A as an attribute.
  - The primary key of R is the combination of A and K. If the multivalued attribute is composite, we include its simple components.
- **Example:** The relation DEPT_LOCATIONS is created.
  - The attribute DLOCATION represents the multivalued attribute LOCATIONS of DEPARTMENT, while DNUMBER-as foreign key-represents the primary key of the DEPARTMENT relation.
  - The primary key of R is the combination of {DNUMBER, DLOCATION}.

ER-to-Relational Mapping Algorithm

Step 6: Mapping of Multi valued attributes.

DEPT_LOCATIONS

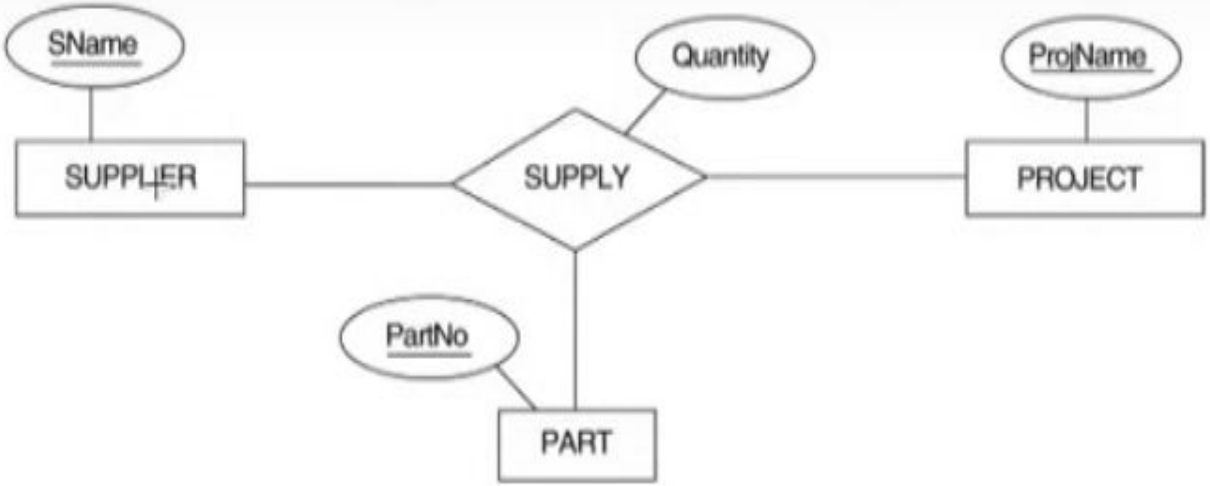
Dnumber	Dlocation



ER-to-Relational Mapping Algorithm

- **Step 7: Mapping of N-ary Relationship Types.**
  - For each n-ary relationship type R, where n>2, create a new relationship S to represent R.
  - Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types.
  - Also include any simple attributes of the n-ary relationship type (or simple components of composite attributes) as attributes of S.
- **Example:** The relationship type SUPPY in the ER on the next slide.
  - This can be mapped to the relation SUPPLY shown in the relational schema, whose primary key is the combination of the three foreign keys {SNAME, PARTNO, PROJNAME}

(a)



SUPPLIER

SNAME	. . .

PROJECT

PROJNAME	. . .

PART

PARTNO	. . .

SUPPLY

SNAME	PROJNAME	PARTNO	QUANTITY

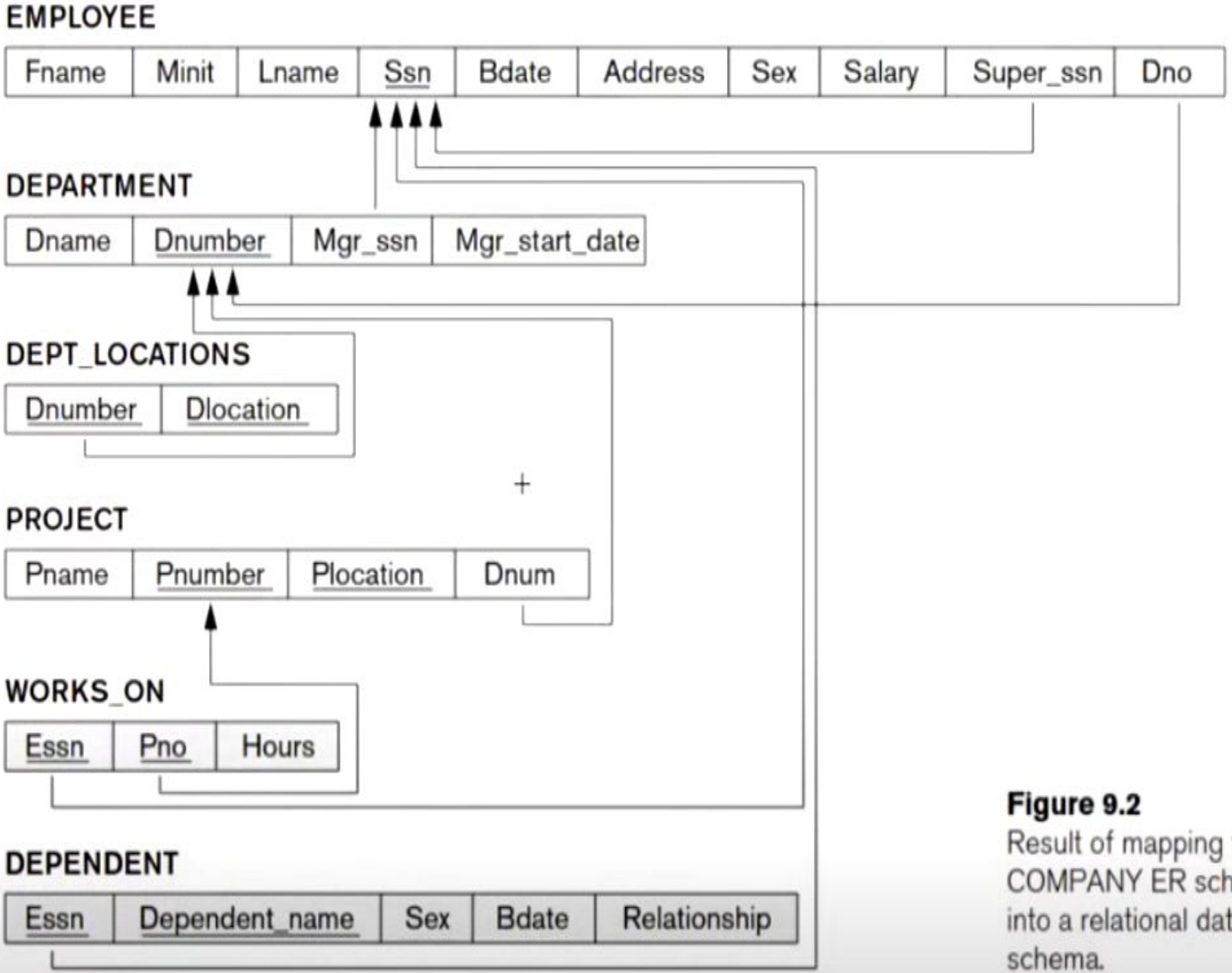


Figure 9.2
Result of mapping the COMPANY ER schema into a relational database schema.